Electroencephalography (EEG)

Electroencephalography (EEG) is an electrophysiological monitoring method to record electrical activity of the brain. It is typically noninvasive, with the electrodes placed along the scalp, although invasive electrodes are sometimes used, as in electrocorticography. EEG measures voltage fluctuations resulting from ionic current within the neurons of the brain. Clinically, EEG refers to the recording of the brain's spontaneous electrical activity over a period of time, as recorded from multiple electrodes placed on the scalp.

In 1920 Hans Berger (Austrian Psychiatrist) was the first to record EEG tracings from human being. He discovered waves at 10 Hz (Alpha waves because they were the first he isolated in human EEG.

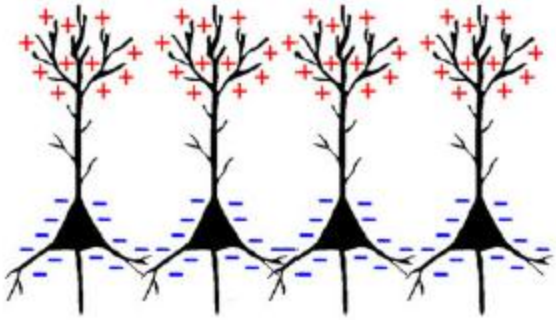
EEG is one of the main diagnostic tests for epilepsy. A routine clinical EEG recording typically lasts 20–30 minutes (plus preparation time). It is a test that detects electrical activity in the brain using small, metal discs (electrodes) attached to the scalp. Routinely, EEG is used in clinical circumstances to determine changes in brain activity that might be useful in diagnosing brain disorders, especially epilepsy or another seizure disorder. An EEG might also be helpful for diagnosing or treating the following disorders-

- ✓ Brain tumor
- ✓ Brain damage from head injury
- ✓ Brain dysfunction that can have a variety of causes (encephalopathy)
- ✓ Inflammation of the brain (encephalitis)
- ✓ Stroke
- ✓ Sleep disorders etc.

Origin of EEG:

Most of what we record is felt to originate from neurons, and there are a number of possible sources including action potentials, post-synaptic potentials (PSPs), and chronic neuronal depolarization. Action potentials induce a brief (10 ms or less) local current in the axon with a very limited potential field. This makes them unlikely candidates. PSPs are considerably longer (50–200 ms), have a much greater field and thus are more likely to be the primary generators of the EEG. Long term depolarization of neurons or even glia could also play a role and produce EEG changes.

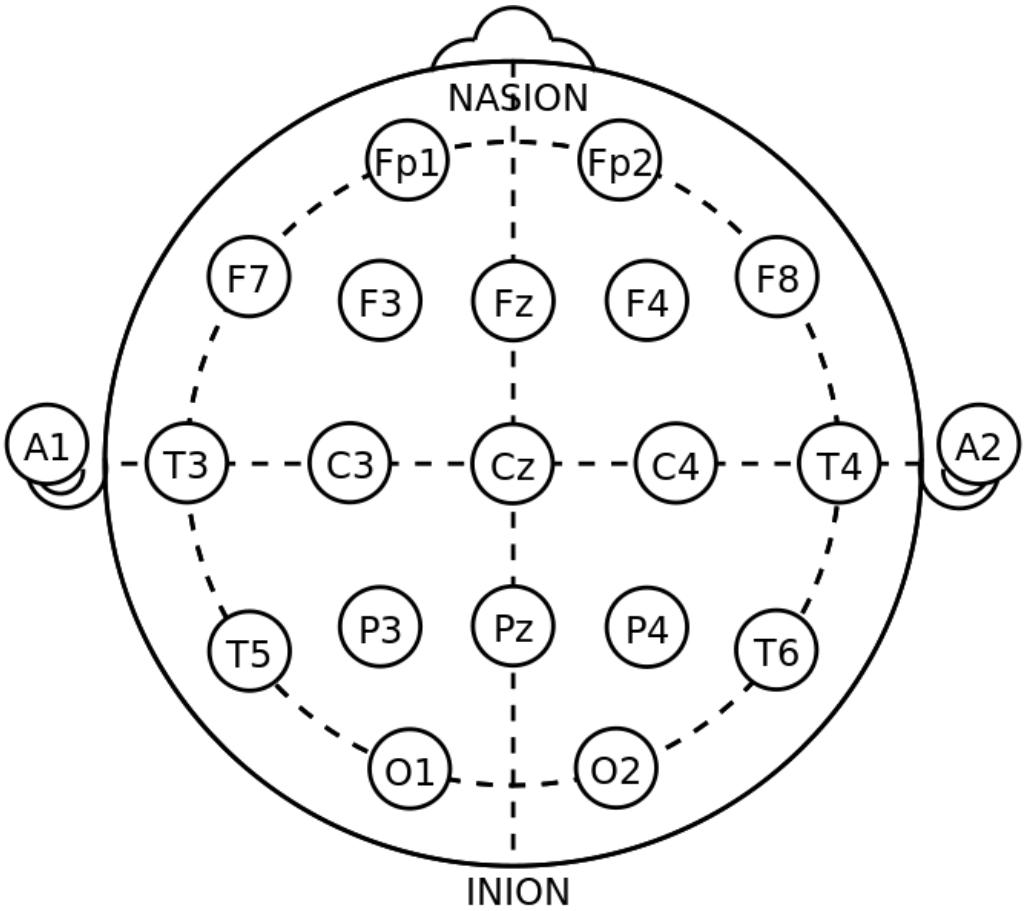
In the normal brain an action potential travels down the axon to the nerve terminal, where a neurotransmitter is released. However, it is the synaptic potentials that are the most important source for the electroencephalogram. The resting membrane potential (electrochemical equilibrium) is typically –70 mV on the inside. At the post-synaptic membrane the neurotransmitter produces a change in membrane conductance and transmembrane potential. If the signal has an excitatory effect on the neuron it leads to a local increase of the transmembrane potential (depolarization) and is called an excitatory post-synaptic potential (EPSP), typically located in the dendrites. Note that during an EPSP the inside of the neuronal membrane becomes more positive while the extracellular matrix becomes more negative. Inhibitory post-synaptic potentials (IPSPs) result in local hyperpolarization typically located on the cell body of the neuron. The combination of EPSPs and IPSPs induces currents that flow within and around the neuron with a potential field sufficient to be recorded on the scalp.



The electric potential generated by an individual neuron is far too small to be picked up by EEG. EEG activity therefore always reflects the summation of the synchronous activity of thousands or millions of neurons that have similar spatial orientation. If the cells do not have similar spatial orientation, their ions do not line up and create waves to be detected. Pyramidal neurons of the cortex are thought to produce the most EEG signal because they are well-aligned and fire together.

10-20 system for electrode placement:

The **10–20 system** or **International 10–20 system** is an internationally recognized method to describe and apply the location of scalp electrodes in the context of an EEG exam. This method was developed to maintain standardized testing methods ensuring that a subject's study outcomes (clinical or research) could be compiled, reproduced, and effectively analyzed and compared using the scientific method. The system is based on the relationship between the location of an electrode and the underlying area of the brain, specifically the cerebral cortex.
The "10" and "20" refer to the fact that the actual distances between adjacent electrodes are either 10% or 20% of the total front–back or right–left distance of the skull.



Electrode labeling:

Each electrode placement site has a letter to identify the lobe, or area of the brain it is reading from: pre-frontal (Fp), frontal (F), temporal (T), parietal (P), occipital (O), and central (C). Note that there is no "central lobe"; due to their placement, and depending on the individual, the "C" electrodes can exhibit/represent EEG activity more typical of frontal, temporal, and some parietal-occipital activity

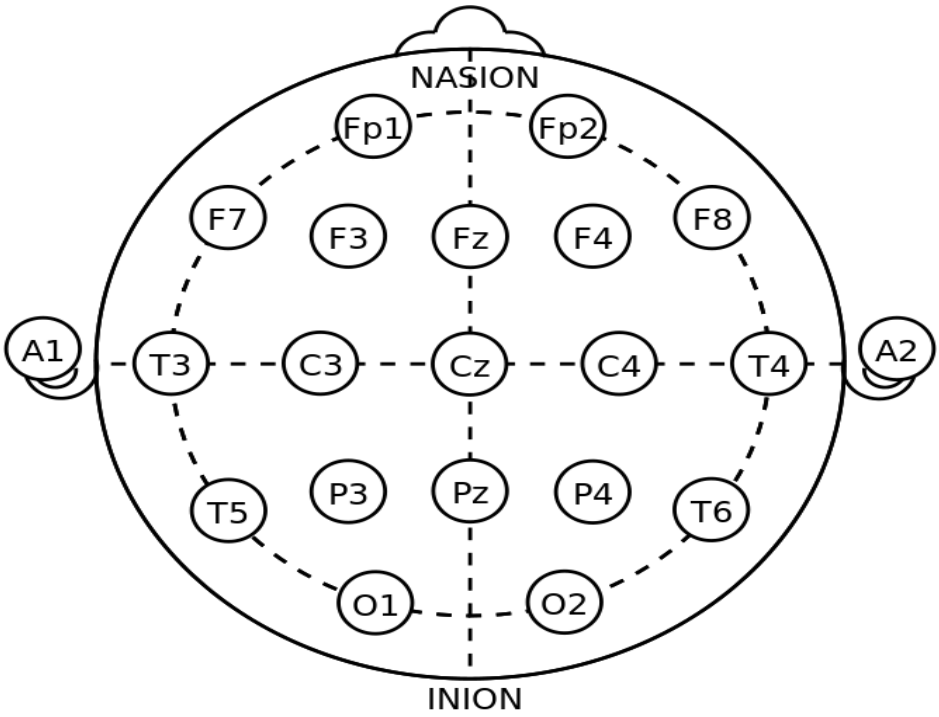
There are also (Z) sites: A "Z" (zero) refers to an electrode placed on the midline sagittal plane of the skull, (FpZ, Fz, Cz, Oz) and is present mostly for reference/measurement points. These electrodes will not necessarily reflect or amplify lateral hemispheric cortical activity as they are placed over the corpus callosum, and do not represent either hemisphere adequately. "Z" electrodes are often utilized as 'grounds' or 'references.

Even-numbered electrodes (2,4,6,8) refer to electrode placement on the right side of the head, whereas odd numbers (1,3,5,7) refer to those on the left.

Measurement:

Specific anatomical landmarks are used for the essential measuring and positioning of the EEG electrodes. These are found with a tape measure, and often marked with a pencil marker.

- ✓ Nasion to Inion (Sagittal Plane): Marks for the Z electrodes are made between these points along the midline, at intervals of 10%, 20%, 20%, 20%, 20% and 10%.
- ✓ Preauricular to preauricular (Coronal Plane): The T3, C3, Cz, C4, and T4 electrodes are placed at marks made at intervals of 10%, 20%, 20%, 20%, 20% and 10%, respectively, measured across the top of the head.
- ✓ Skull circumference is measured just above the ears (T3 and T4), just above the bridge of the nose (at Fpz), and just above the occipital point (at Oz). The Fp2, F8, T4, T6, and O2 electrodes are placed at intervals of 5%, 10%, 10%, 10%, 10%, and 5%, respectively, measured above the right ear, from front (Fpz) to back (Oz). The same is done for the odd-numbered electrodes on the left side, to complete the full circumference.
- ✓ The Fp1, F3, C3, P3, and O1 electrodes are placed at marks made at intervals of 25%, 25%, 25%, 25% respectively, measured across the right parasagittal plane from front to back. The Fp2, F4, C4, P4, and O2 electrodes are also placed at the same manner on the left side.

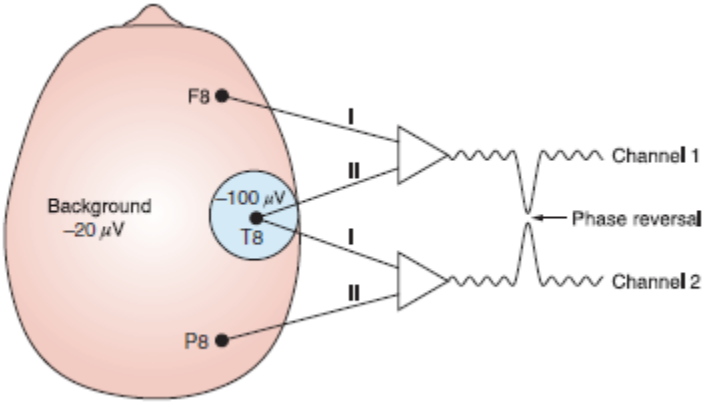


EEG Recording:

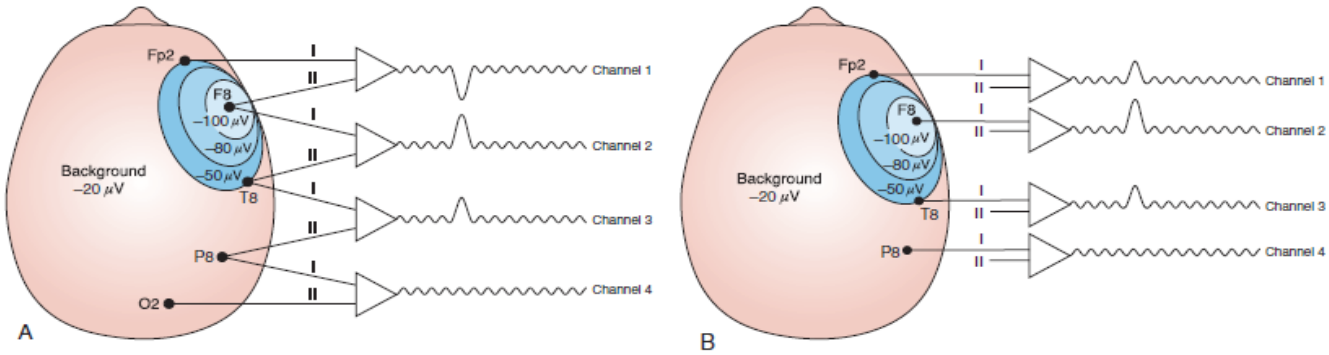
Bipolar Recording: Bipolar recordings electronically link successive electrodes (known as a chain or line). The voltage at one electrode is compared with the voltage affecting adjacent electrodes (potential difference). Each amplifier has two inputs, I and II. By convention, the rules for understanding the display are:

- ✓ If input I becomes negative with respect to input II, there is an upward deflection.
- ✓ If input II becomes negative with respect to input I, there is a downward deflection.

Contrary to our well-used Cartesian coordinate system, the convention in neurophysiology is that an upward deflection is negative and a downward deflection is positive.



Referential Recording: In referential recording the amplifiers are not linked as in bipolar recording. Signals from each of the scalp electrodes are conducted to input I of the associated amplifier, while signals from the reference are conducted to input II. Thus, in referential recording, we record the potential difference between a particular scalp electrode and a referential electrode. Reference montages produce a higher amplitude EEG recording because of the longer inter electrode distances. Theoretically, the reference can be located anywhere, but there are practical considerations. A reference placed at any distant point will be contaminated with ambient electrical noise, 60 Hz artifact (50 Hz in Europe). A reference placed on, say, the shoulder or chest would also pick up high-voltage EKG artifact. Interference from an EKG would render the EEG unreadable. The ears are relatively free from both these artifacts, so ear is used as reference.



A=Bipolar recording & B= Reference recording

Common Average Recording: A widely used reference is the common average reference. In this scheme, the voltage of an event occurring under a particular electrode (input I) is compared with the average voltage recorded by all the electrodes on the scalp (input II).

EEG montages: Montage refers to the pattern of systematic linkage of the scalp electrodes designed to obtain a logical display of the electrical activity.

- ✓ **Bipolar Longitudinal montage:** Adjacent electrodes are connected from front to back, including the temporal (lateral) chain and the parasagittal chain. The EEG is displayed in various ways. In this example, the four channels of the temporal chain on one side are followed by the temporal channels on the opposite side. Similarly, the four channels of the parasagittal chain also alternate.

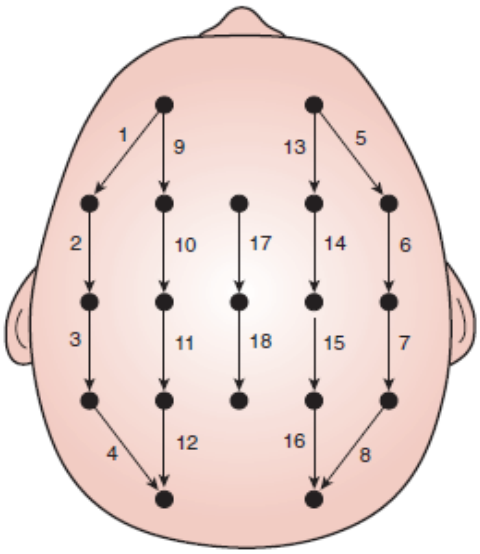
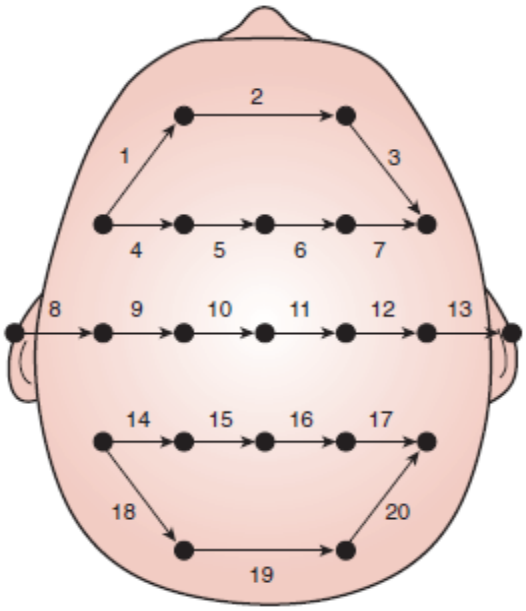


Figure shows a typical longitudinal bipolar montage. The numbers refer to channels, reflecting voltage difference between two electrodes. Both temporal and parasagittal chains alternate from left to right. The arrows represent the inputs to each amplifier.

- ✓ **Bipolar Transverse montage:** A second popular arrangement is the transverse bipolar montage. This links adjacent electrodes in transverse chains, starting anteriorly and progressing posteriorly. Each chain starts with the left side and progresses to the right.

Figure shows a transverse bipolar montage. The chains run from left to right, beginning anteriorly and proceeding posteriorly. Note that the midline electrodes are incorporated into the second, third, and fourth chains, thus allowing good representation of midline events.



- ✓ **Bipolar Circumferential Montage:** As the name implies, the circumferential montage encircles the head and is particularly useful for examining spikes and sharp waves, which occur at the end of the longitudinal bipolar chain.

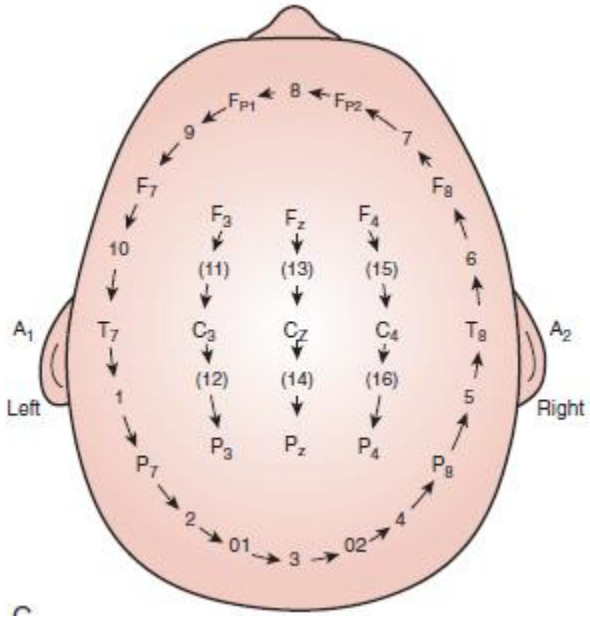


Figure shows Circumferential Bipolar Montage.

EEG (Electroencephalography) waves represent electrical activity in the brain. These waves are classified based on their frequency, and each type is associated with different states of brain activity. Here are the primary EEG wave types:

1. **Delta Waves** (0.5 - 4 Hz):
   - **Characteristics**: High amplitude, slow frequency.
   - **Associated States**: Deep sleep, unconsciousness, or very relaxed states.
   - **Use**: Often observed in stages 3 and 4 of deep sleep.
2. **Theta Waves** (4 - 8 Hz):
   - **Characteristics**: Slow frequency, moderate amplitude.
   - **Associated States**: Drowsiness, light sleep, deep relaxation, meditative states, and creative thinking.
   - **Use**: Common in light sleep (Stage 1) or during deep relaxation and meditation.
3. **Alpha Waves** (8 - 12 Hz):
   - **Characteristics**: Moderate amplitude, rhythmical waves.
   - **Associated States**: Calm, relaxed, but alert states (e.g., when awake with eyes closed).
   - **Use**: Often seen in relaxed wakefulness, especially with closed eyes.
4. **Beta Waves** (12 - 30 Hz):
   - **Characteristics**: High frequency, low amplitude.
   - **Associated States**: Active thinking, alertness, problem-solving, focus, and anxiety.
   - **Use**: Common in states of concentration and mental activity.
5. **Gamma Waves** (30 - 100 Hz):
   - **Characteristics**: Highest frequency, low amplitude.
   - **Associated States**: Higher mental activity, perception, memory processing, and cognitive functioning.
   - **Use**: Observed during high-level cognitive functions, including learning, attention, and memory processing.

Summary of EEG Waves:

- **Delta**: Deep sleep, unconsciousness.
- **Theta**: Drowsiness, relaxation, creativity.
- **Alpha**: Calm and relaxed wakefulness.
- **Beta**: Focused attention, anxiety, active thinking.
- **Gamma**: High-level cognitive processing, memory.

EEG waves help in diagnosing various neurological conditions, including epilepsy, sleep disorders, and cognitive function impairments.